PRIYANSHU

AI & Systems Engineer | Indian Institute of Technology Kharagpur

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2024 – 2028	B.S. (Hons) in Applied Geology	IIT Kharagpur	8.47 / 10.0
2024	CBSE Class XII (Senior Secondary)	R R S Sr. Sec. School, Gurgaon	89.2%
2022	ICSE Class X (Secondary)	Lipin Public School, Bhiwadi	96.8%

EXPERIENCE

Software Engineering Intern (AI & Backend) | **Ozric Technologies (OPC) Pvt. Ltd.** May 2026 – Jun 2026

- **Production RAG Pipeline:** Architected an end-to-end RAG system using dense embeddings and Qdrant/pgvector to index domain knowledge, applying semantic chunking and hybrid retrieval to ground LLM responses with source citations.
- **Context Engineering:** Implemented dynamic prompt context assembly with sliding-window token budgeting and cross-encoder re-ranking, reducing token overhead by 35% and cutting hallucination rates across multi-turn queries.
- **Scalable Backend Microservices:** Built high-throughput asynchronous REST APIs using FastAPI, Pydantic, and SQLAlchemy ORM on PostgreSQL with Alembic migrations, supporting multi-tenant queries with sub-150ms p95 latency.
- **Security & Delivery:** Enforced tenant data isolation via JWT and Role-Based Access Control (RBAC); containerized services with Docker and automated testing via Pytest, achieving 90%+ integration test coverage.

PROJECTS

NERA: AI-Based Smart Logistics and Accessibility Intelligence Platform | GitHub SIH 2026

- **Multimodal Hazard Modeling:** Engineered an AI + GIS decision-support platform (SIH 2026), combining real-time meteorological feeds, SRTM terrain elevation models, and road-network topologies for North-East transit corridors.
- **ML Inference & Custom Routing Engine:** Trained an XGBoost hazard classifier served via FastAPI, integrated with a custom Go graph engine (A*/Dijkstra) over in-memory OpenStreetMap (OSM) data for dynamic route re-calculation.
- **Full-Stack Spatial Visualization:** Built a React/TypeScript Mapbox dashboard for live hazard mapping and route inspection; secured a **Top-10 ranking** among 300+ competing teams at IIT Kharagpur.

MiniContainer + Container Observability | **Linux Systems in C++** | Runtime | Observability Jan 2026 – Mar 2026

- **Kernel-Level Containerization:** Built a lightweight Linux container runtime in C++ using namespaces (`clone` flags: PID, Mount, UTS, Net), `pivot_root`, and cgroups v2 controllers (`cpu.weight`, `memory.max`) for rigorous sandboxed execution.
- **Low-Overhead Telemetry Daemon:** Developed a companion C++ observability daemon directly reading kernel pseudo-filesystems (`/sys/fs/cgroup`) to stream live container discovery, CPU, memory, and process metrics with zero overhead.
- **Benchmarking & Systems Depth:** Achieved ∼**7 ms** cold-start latency vs 506 ms for Docker (∼**72**× **lower startup overhead**), establishing optimal low-latency isolation for serverless AI inference tasks and code-execution sandboxes.

Network Security: End-to-End Phishing Detection Platform | GitHub Nov 2025 – Dec 2025

- **Production MLOps Pipeline:** Engineered an end-to-end classification pipeline with automated data ingestion, validation, and lexical/DNS feature extraction, achieving a **99.1% F1-score** and **98.8% precision**.
- **Data Drift & Experiment Tracking:** Integrated automated covariate drift detection using two-sample Kolmogorov-Smirnov (KS) tests; tracked hyperparameter sweeps and model artifacts via MLflow and DagsHub for automated retraining.
- **Containerized Inference Service:** Packaged model into an asynchronous FastAPI service containerized with Docker, integrating automated regression and linting suites via GitHub Actions CI/CD.

OPEN-SOURCE CONTRIBUTIONS

- **Prometheus Operator (CNCF)** | PR #8813: Fixed critical configuration drift where unsupported Prometheus `remote_write` spec fields propagated to Thanos components; authored Go filtering logic within the reconciliation engine to sanitize YAML specs, eliminating Thanos pod crash-loops across Kubernetes telemetry clusters (Merged, 22/22 CI passed).
- **AsyncAPI (Linux Foundation)** | PR #77: Diagnosed schema traversal and deserialization failures under the AsyncAPI v3 event specification; engineered recursive property-rendering logic to accurately resolve nested object schemas, `$ref` definitions, and dynamic message payloads (Merged).
- **OneBusAway/Maglev (Go Transit Engine)** | PRs #1428, #1375, #1378: Authored multiple merged fixes in Go: resolved cumulative block distance calculation errors across transitions (`distanceAlongBlock`), repaired agency references, and eliminated swallowed errors in concurrent trip lookups.

SKILLS

- **Languages:** Python, C++, Go, SQL, C, JavaScript/TypeScript, Bash
- **GenAI & LLMs:** RAG Systems, Vector Databases (Qdrant, pgvector), Dense Embeddings, Semantic Search, Prompt Engineering, Context Budgeting, Hallucination Mitigation, vLLM, Ollama
- **Machine Learning & MLOps:** XGBoost, Scikit-Learn, PyTorch, Feature Engineering, Kolmogorov-Smirnov Drift Detection, MLflow, DagsHub, Model Evaluation (F1, Precision-Recall, AUC-ROC), NumPy, Pandas
- **Backend, Cloud & Observability:** FastAPI, Asynchronous REST APIs, Pydantic, PostgreSQL, MongoDB, Docker, Git, CI/CD
- **Computer Science Core:** DSA, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming (OOP)

COURSEWORK AND LEADERSHIP

Coursework: Programming and Data Structures (Theory & Lab), Essentials of Machine Learning, Probability and Statistics, Linear Algebra, Advanced Calculus.

Secretary, Games and Sports Society, Azad Hall of Residence (2025–2026): Directed hall-wide athletic events and administered an operational budget exceeding Rs. 1,00,000; Silver Medalist, GC Men's Senior Hammer Throw.

Open IIT Product Management Competition 2025, Gold Medal: Designed architecture of an AI-agent-driven ERP.